

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

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COURSE NAME: Deep Learning
FACULTY : Dr. Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

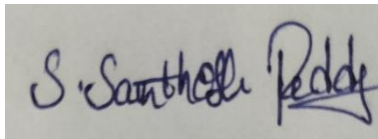
Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code Of Conduct:

I **S. Santhosh Reddy (192425017)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

S. Santhosh Reddy
Name of the student:

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

Answer :

1. Learning Algorithms (Optimization Behavior)

High initial learning rate with decay:

- **Effect on convergence:**
 - A high initial learning rate allows the optimizer to make large jumps, escaping shallow local minima and quickly exploring the parameter space.
 - Gradual decay ensures finer adjustments later, stabilizing convergence near the global minimum.
- **When it outperforms constant learning rate:**
 - In convex problems, this strategy can converge faster than a small constant rate, which may be too slow.
 - It avoids the risk of stagnation in flat regions by starting aggressively, then refining.
- **Risks:**
 - Too high a rate may cause divergence or oscillations around the minimum.
 - If decay is too fast, the optimizer may stop improving prematurely; if too slow, instability persists.
 - Requires careful tuning of initial rate and decay schedule.

2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean μ and known variance σ^2

Question:

Derive the Maximum Likelihood Estimator for μ , and analytically explain how the estimate changes when:

- The dataset contains outliers
 - The sample size increases
- What does this imply about the robustness of MLE?

Answer:

Maximum Likelihood Estimation (MLE)

Gaussian with unknown mean μ , known variance σ^2 :

- **MLE derivation:** The likelihood is

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Log-likelihood:

$$\ell(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Differentiating and setting derivative to zero:

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i$$

→ The sample mean.

- **With outliers:** The mean shifts significantly, since MLE is sensitive to extreme values.
- **With larger sample size:** The influence of individual outliers diminishes, and the estimate converges to the true mean (law of large numbers).
- **Implication:** MLE is efficient but not robust; it has low variance with large samples but poor resistance to outliers.

3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

Answer:

Building a Machine Learning Algorithm (Model Selection) 10,000 features, 500 samples:

- **Challenges:**
 - High dimensionality → risk of overfitting.
 - Small sample size → poor generalization.
 - Computational inefficiency.
- **Proposed pipeline:**
 1. **Feature selection / dimensionality reduction:**
 - Use PCA or autoencoders to reduce dimensionality.
 - Alternatively, apply statistical tests or mutual information to select relevant features.
 2. **Regularization:**
 - Use L1 (Lasso) to enforce sparsity, or L2 (Ridge) to penalize large weights.
 3. **Cross-validation:**
 - Employ k-fold CV to maximize use of limited data.
 4. **Model choice:**
 - Linear models or kernel methods with regularization are preferable over complex deep nets.
- **Mitigation of overfitting:**
 - Dimensionality reduction lowers variance.
 - Regularization prevents overly complex models.
 - Cross-validation ensures stability of results.

4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

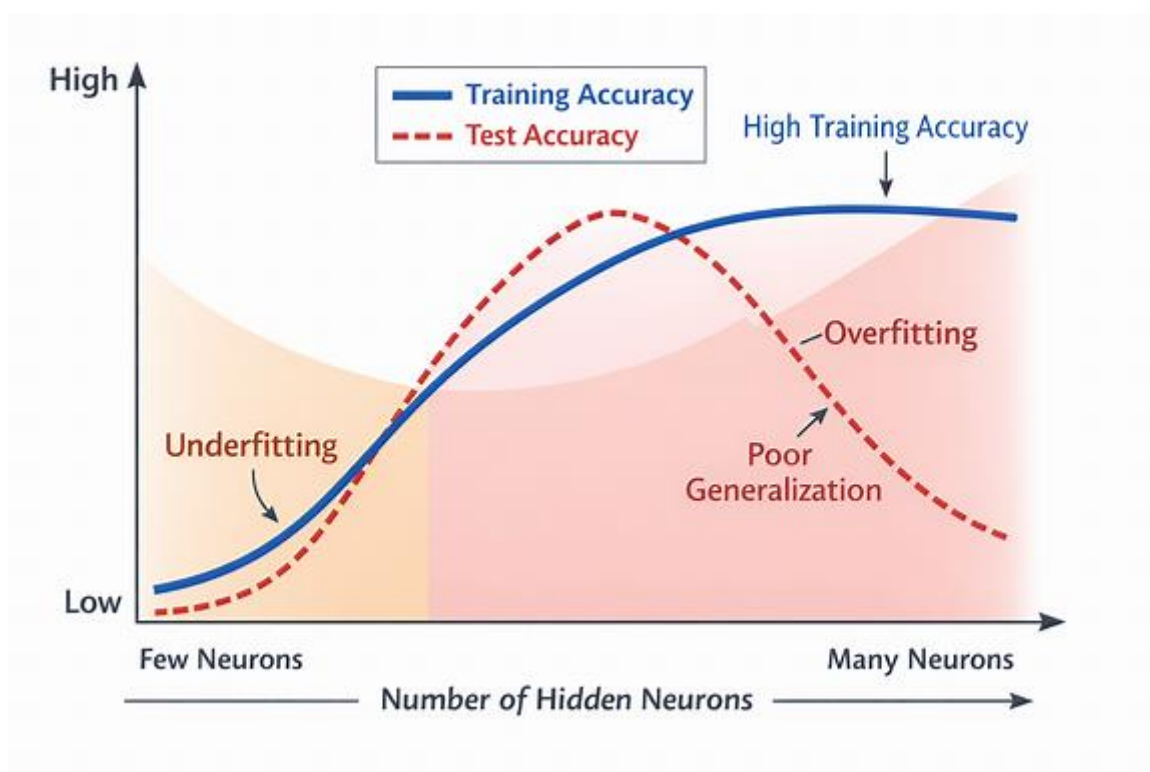
Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

Answer:

Neural Networks (MLP with sigmoid activations)

- **Adding hidden neurons:**
 - Increases representational power: more neurons allow approximation of more complex functions (Universal Approximation Theorem).
 - Each neuron adds nonlinear transformations, expanding the hypothesis space.
- **Why not always better:**
 - **Overfitting:** Too many neurons memorize training data instead of generalizing.
 - **Vanishing gradients:** Sigmoid activations saturate, slowing learning.
 - **Generalization trade-off:** More capacity requires more data; with limited data, performance may degrade.
- **Conclusion:** More neurons expand potential but must be balanced with regularization and sufficient data.



5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Propose at least two techniques to overcome these issues and justify them analytically.

Answer:

Backpropagation, SGD & Curse of Dimensionality

Impact of high dimensionality:

- **Gradient updates:**
 - Gradients become noisy and less informative; many irrelevant directions dilute signal.
- **Convergence speed:**
 - SGD struggles as effective step sizes shrink in sparse, high-dimensional spaces.
 - More iterations are needed to find meaningful directions.
- **Generalization:**
 - Models risk overfitting noise in high-dimensional data.
 - True structure is harder to capture.
- **Techniques to overcome:**
 1. **Dimensionality reduction (PCA, autoencoders):** Removes redundant features, focusing learning on informative subspaces.
 2. **Regularization (Dropout, L1/L2 penalties):** Prevents overfitting by constraining complexity.
 3. **Mini-batch normalization / adaptive optimizers (Adam, RMSProp):** Stabilize gradient updates in high dimensions.
- **Analytical justification:**
 - Reducing dimensionality lowers variance and improves gradient signal.
 - Regularization balances bias-variance trade-off.
 - Adaptive methods adjust learning rates per dimension, mitigating sparse gradient issues.